



Formaldehyde scavengers for cleaner production: A case study focused on the leather industry

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ARTICLE INFO

Article history:

Available online 12 March 2018

Handling Editor: Cecilia Maria Villas Bôas de Almeida

Keywords:

Cleaner production

Leather

Textile

Wood industries

Formaldehyde resins

Scavengers

Formaldehyde content reduction

ABSTRACT

Due to its carcinogenic character, the presence of formaldehyde in leather continues to be a subject of great concern. By using formaldehyde scavengers, it is possible to reduce the formaldehyde content in leather. In this work, the potential ability of three different compounds (ethylene urea, pyrogallol and gallic acid) to reduce the formaldehyde content in splits leathers treated with formaldehyde resins (melamine-formaldehyde and dicyandiamide-formaldehyde) is assessed. This capacity is compared with that of a fourth scavenger (hydroxylamine sulphate) already used in tanneries. The evolution of the formaldehyde content with time is also considered, as well as the potential coadjuvant effect of other compounds such as mimosa extract and an acid dye (Acid Black 234). Hydroxylamine sulphate initially showed the highest ability to reduce formaldehyde content. However, after a certain time, this ability proved to be inferior to the ability of other compounds due to the reversibility of the reaction between hydroxylamine and formaldehyde. Pyrogallol showed a higher ability than gallic acid when used in the final wash of leather processing. However, the treatment with pyrogallol results in a darkening of the leather; this darkening limits its use. Gallic acid may be a good alternative to formic acid as the final fixing agent in leather processing when the presence of formaldehyde in leathers is suspected. The use of gallic acid in the final wash or as a fixing agent fulfils the formaldehyde content limit (65–75 mg/kg) of the major brands in leather goods in direct contact with the skin. The addition of 2% of gallic acid in the final wash of leather processing resulted in formaldehyde content reductions that varied from 65% to 85%. However, further experiments are required to assess the influence of gallic acid on the fastness properties and the coloration acquired by the treated leathers. The joint effect of gallic acid in the final wash or as a fixing agent and mimosa extract as a retanning agent in formaldehyde content reduction is even enhanced by subsequently using a dye with amino groups in its chemical structure. Reducing the formaldehyde content by using scavengers can contribute to the achievement of a cleaner production in those sectors (leather, textile, wood) that use formaldehyde resins.

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1. Introduction

The analysis of future trends in leather production emphasizes the introduction of cleaner leather processing technologies (UNIDO, 2010). One of the expected results includes avoiding the presence in the leather of substances from the Restricted Substances Lists (RSL) due to its proven impact on human health and ecosystems (UNIDO Leather Panel, 2010) as well as its use in leather

processing. Due to the carcinogenic character of formaldehyde (IARC Monographs, 2012; Dixit et al., 2015), its presence in leather continues to be a subject of great concern.

The worldwide production of leather for footwear in 2014 amounts to 561.1 thousand tonnes (FAO, 2016). An estimation of approximately 60%–70% of this production is retanned with resins synthesized with formaldehyde and compounds such as melamine, dicyandiamide and urea. Therefore, a remarkable amount of leather for footwear may pose formaldehyde problems.

According to the results given by the “A3 Chair in Leather Innovation” since 2013 on the formaldehyde content in leather for footwear from Asia, 16% of leathers failed the requirements of most

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